

Chapter 2

SETS, FUNCTIONS AND SEQUENCES

EXERCISE 2.1

Multiple Choice Questions (MCQs)

Q.1 Tick ✓ the correct option in the following questions:

(i) If A and B are sets, then the logical equivalence of $A = B$ is

- (a) $\forall x(x \in A \rightarrow x \in B)$ (b) $\forall x(x \in B \rightarrow x \in A)$
(c) $\exists x(x \in A \leftrightarrow x \in B)$ (d) $\forall x(x \in A \leftrightarrow x \in B)$

Sol: Correct answer is option (d).

(ii) If A and B are sets, then the logical equivalence of $A \subseteq B$ is

- (a) $\forall x(x \in A \rightarrow x \in B)$ (b) $\forall x(x \in B \rightarrow x \in A)$
(c) $\exists x(x \in A \leftrightarrow x \in B)$ (d) $\forall x(x \in A \leftrightarrow x \in B)$

Sol: Correct answer is option (a).

(iii) If A and B are sets, then the logical equivalence of $B \subseteq A$ is

- (a) $\forall x(x \in A \rightarrow x \in B)$ (b) $\forall x(x \in B \rightarrow x \in A)$
(c) $\exists x(x \in A \leftrightarrow x \in B)$ (d) $\forall x(x \in A \leftrightarrow x \in B)$

Sol: Correct answer is option (b).

(iv) Which of the following is not a singleton set?

- (a) $\{\{1, 2\}\}$ (b) $\{1, 2\}$ (c) $\{\emptyset\}$ (d) $\{\emptyset, 1\}$

Sol: Correct answer is option (d).

(v) The logical statement $\forall x(x \in \emptyset \rightarrow x \in S)$ is equivalent to

- (a) $S \subseteq \emptyset$ (b) $\emptyset \not\subseteq S$ (c) $\emptyset \subseteq S$ (d) $\emptyset = S$

Sol: Correct answer is option (c).

(vi) The logical statement $\forall x(x \in \emptyset \leftrightarrow x \in S)$ is equivalent to

- (a) $S \subseteq \emptyset$ (b) $\emptyset \not\subseteq S$ (c) $\emptyset \subseteq S$ (d) $\emptyset = S$

Sol: Correct answer is option (d).

(vii) If A is a proper subset of B , then logical statement is

- (a) $\forall x(x \in A \rightarrow x \in B) \wedge \exists x(x \in B \wedge x \notin A)$
 (b) $\forall x(x \in A \rightarrow x \in B) \wedge \exists x(x \in B \wedge x \notin A)$
 (c) $\forall x(x \in A \rightarrow x \in B) \vee \exists x(x \in B \wedge x \notin A)$
 (d) $\forall x(x \in A \rightarrow x \in B) \vee \exists x(x \in B \wedge x \in A)$

Sol: Correct answer is option (b).

(viii) if $A = \{1, 2, 3, 4\}$, then $|A| =$

- (a) 1 (b) 2 (c) 3 (d) 4

Sol: Correct answer is option (d).

(ix) Cardinality of an empty set is

- (a) -2 (b) -1 (c) 0 (d) 1

Sol: Correct answer is option (c).

(x) Number of elements in the power set of $A = \{1, 2, 3\}$ is

- (a) 2 (b) 3 (c) 6 (d) 8

Sol: Correct answer is option (d).

(xi) If $R = \{(x, y) : x = y\}$ is a relation on $\{1, 2, 3\}$, then $R =$

- (a) $\{(1, 1), (2, 2)\}$ (b) $\{(2, 2), (3, 3)\}$
 (c) $\{(1, 1), (3, 3)\}$ (d) $\{(1, 1), (2, 2), (3, 3)\}$

Sol: Correct answer is option (d).

Short Questions

Solve / write answers of the following short questions:

Q.2 List the members of these sets.

(i) $\{x : x \text{ is a positive integer such that } x^2 = 1\}$

Sol: $\{1\}$

(ii) $\{x : x \text{ is a real number such that } x^2 = 1\}$

Sol: $\{-1, 1\}$

(iii) $\{x : x \text{ is a real number such that } x^2 = 1\}$

Sol: $\{1\}$

(iv) $\{x : x \text{ is a positive integer less than 15}\}$

Sol: $\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14\}$

(v) $\{x : x \text{ is the square of an integer and } x < 100\}$

Sol: $\{0, 1, 4, 9, 16, 25, 36, 49, 64, 81\}$

(vi) $\{x : x \text{ is an integer such that } x^2 = 2\}$

Sol: ϕ

(vii) $\{x : x \text{ is an integer such that } x^2 = 7\}$

Sol: ϕ

(viii) $\{x : x \text{ is the cube of a positive integer and } x < 100\}$

Sol: $\{1, 8, 27, 64\}$

(ix) $\{x : x \in \mathbb{R} \wedge 0 < x < 1\}$

Sol: $]0, 1[$

(x) $\{x : x \in \mathbb{R} \wedge -1 < x < 1\}$

Sol: $] -1, 1[$

Q.3 Use set builder notation to give a description of each of these sets.

(i) $\{0, 2, 4, 6, 8, 10, 12, 14\}$

Sol: $\{x : x \text{ is a multiple of } 2 \wedge 0 \leq x \leq 14\}$

(ii) $\{4, 8, 12, 16, 20, 24, 28, 32\}$

Sol: $\{x : x \text{ is a multiple of } 4 \wedge 4 \leq x \leq 32\}$

(iii) $\{x : x \text{ is a real number such that } x^3 = 1\}$

Sol: $\{x : x \text{ is a real number such that } x^3 = 1\}$

(iv) $\{-3, -2, -1, 0, 1, 2, 3\}$

Sol: $\{x : x \text{ is an integer } \wedge -3 \leq x \leq 3\}$

(v) $\{-1, 0, 1\}$

Sol: $\{x : x \text{ is an integer } \wedge -1 \leq x \leq 1\}$

Q.4 Find the truth set of each of these predicates where the domain is the set of integers.

(i) $P(x) : x^3 \geq 1$

Sol: Set of natural numbers

(ii) $P(x) : x^2 < 3$

Sol: $\{-1, 0, 1\}$

(iii) $Q(x) : x^2 > x$

Sol: Set of integers

(iv) $\{-3, -2, -1, 0, 1, 2, 3\}$

Sol: $\{-3, -2, -1, 0, 1, 2, 3\}$

(v) $R(x) : 2x + 1 = 0$

Sol: $2x + 1 = 0$ | $x = -\frac{1}{2}$
 $2x = -1$

Since $-\frac{1}{2}$ is not an integer, so the truth set of $R(x) : 2x + 1 = 0$ is

an empty set, i.e. ϕ .

(vi) $R(x) : x < x^2$

Sol: Set of integers

(vii) $Q(x) : x^2 = 2$

Sol: $x^2 = 2 \Rightarrow x = \pm\sqrt{2}$

Since $\pm\sqrt{2}$ are not integers, so the truth set of $Q(x) : x^2 = 2$ is an empty set, i.e. ϕ .

Q.5 Translate each of these quantifications into English and determine its truth value.

(i) $\exists x \in \mathbb{Z} (x + 1 > x)$

(ii) $\exists x \in \mathbb{R} (x^2 = x)$

(iii) $\forall x \in \mathbb{Z} (x^2 \in \mathbb{Z})$

(iv) $\exists x \in \mathbb{R} (x^3 = -1)$

(v) $\forall x \in \mathbb{Z} (x - 1 \in \mathbb{Z})$

(vi) $\forall x \in \mathbb{R} (x^2 \neq -1)$

(vii) $\forall x \in \mathbb{Z} (x^2 > 0)$

(viii) $\exists x \in \mathbb{Z} (x^2 = 2)$

Sol:(i) Some integer added one is greater than integer.

Truth set : Set of integers

(ii) Square of some real number is equal to real number itself.

- Truth set : $\{-1, 0, 1\}$
- (iii) Square of every integer is an integer.
Truth set : Set of integers
- (iv) Cube of some real number is -1 .
Truth set : $\{-1\}$
- (v) Integer minus one is always an integer.
Truth set : Set of integers
- (vi) Square of no real number is -1 .
Truth set : Set of real numbers
- (vii) Square of integer is positive.
Truth set : $\{\pm 1, \pm 2, \pm 3, \dots\}$
- (viii) There is an integer whose square is 2. Truth set : $\emptyset$
- Q.6** Determine whether each of these statements is true or false.

- | | |
|---|--------------------------------------|
| (i) $0 \in \emptyset$ | (ii) $\emptyset \in \{0\}$ |
| (iii) $\emptyset \subset \{0\}$ | (iv) $\{0\} \subset \emptyset$ |
| (v) $\{0\} \in \{0\}$ | (vi) $\{0\} \subset \{0\}$ |
| (vii) $\{\emptyset\} \subseteq \{\emptyset\}$ | (viii) $\emptyset \in \{\emptyset\}$ |
| (ix) $\emptyset \in \{\emptyset, \{\emptyset\}\}$ | (x) $x \in \{x\}$ |
| (xi) $\emptyset \in \{x\}$ | (xii) $\{x\} \in \{x\}$ |
| (xiii) $\emptyset \subseteq \{x\}$ | (xiv) $\{x\} \in \{\{x\}\}$ |
| (xv) $\{x\} \subseteq \{x\}$ | |
-
- | | | | |
|-----------------------|------------|------------|-------------|
| Sol: (i) False | (ii) False | (iii) True | (iv) False |
| (v) False | (vi) False | (vii) True | (viii) True |
| (ix) True | (x) True | (xi) False | (xii) False |
| (xiii) True | (xiv) True | (xv) True | |

Q.7 Solve / write answers of the following short questions:

- (i) Find A^2 if $A = \{0, 1\}$.
Sol: $A^2 = \{(0, 0), (0, 1), (1, 0), (1, 1)\}$
- (ii) Find A^2 if $A = \{0, 1, 3\}$.
Sol: $A^2 = \{(0, 0), (0, 1), (0, 3), (1, 0), (1, 1), (1, 3), (2, 0), (2, 1), (2, 3)\}$
- (iii) Find A^2 if $A = \{a, b, c\}$.
Sol: $A^2 = \{(a, a), (a, b), (a, c), (b, a), (b, b), (b, c), (c, a), (c, b), (c, c)\}$
- (iv) Find A^2 if $A = \{1, 2, a, b\}$.
Sol: $A^2 = \{(1, 1), (1, 2), (1, a), (1, b), (2, 1), (2, 2), (2, a), (2, b), (a, 1), (a, 2), (a, a), (a, b), (b, 1), (b, 2), (b, a), (b, b)\}$
- (v) Find A^3 if $A = \{a\}$.
Sol: $A^3 = \{(a, a, a)\}$
- (vi) Find A^3 if $A = \{0, a\}$.
Sol: $A^3 = \{(0, 0, 0), (0, 0, a), (0, a, 0), (a, 0, 0), (a, a, a), (a, a, 0), (a, 0, a), (0, a, a)\}$
- (vii) Find two sets A and B such that $A \in B$ and $A \subseteq B$.
Sol: $A = \{1\}$, $B = \{1, \{1\}\}$
- (viii) What is the cardinality of each of these sets?
- | | |
|--------------------|----------------------------------|
| (a) $\{1\}$ | (b) $\{\{1\}\}$ |
| (c) $\{1, \{1\}\}$ | (d) $\{1, \{1\}, \{1, \{1\}\}\}$ |

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- Sol:** (a) 1 (b) 1
 (c) 2 (d) 3
- (ix) What is the cardinality of each of these sets?
 (a) $\emptyset$ (b) $\{\emptyset\}$
 (c) $\{\emptyset, \{\emptyset\}\}$ (d) $\{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}\}$
- Sol:** (a) 0 (b) 1
 (c) 2 (d) 3

(x) Find the power sets of the following sets, where a and b are distinct elements.

- (a) $\{a\}$ (b) $\{a, b\}$ (c) $\{\emptyset, \{\emptyset\}\}$
- Sol:** (a) $P(\{a\}) = \{\emptyset, \{a\}\}$
 (b) $P(\{a, b\}) = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}$
 (c) $P(\{\emptyset, \{\emptyset\}\}) = \{\emptyset, \{\emptyset\}, \{\{\emptyset\}\}, \{\emptyset, \{\emptyset\}\}\}$

Long Questions

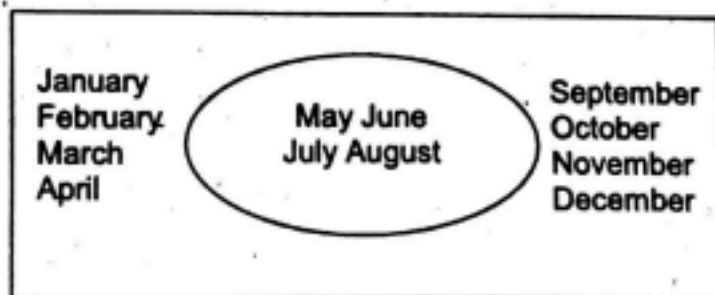
Q.8 Let $A = \{a, b, c\}$, $B = \{x, y\}$, and $C = \{1, 2\}$. Find

- (a) $A \times B \times C$ (b) $C \times B \times A$
 (c) $C \times A \times B$ (d) $B \times B \times B$

Sol: (a) $A \times B \times C = \{(a, x, 1), (a, y, 1), (a, x, 2), (a, y, 2),$
 $(b, x, 1), (b, y, 1), (b, x, 2), (b, y, 2),$
 $(c, x, 1), (c, y, 1), (c, x, 2), (c, y, 2)\}$

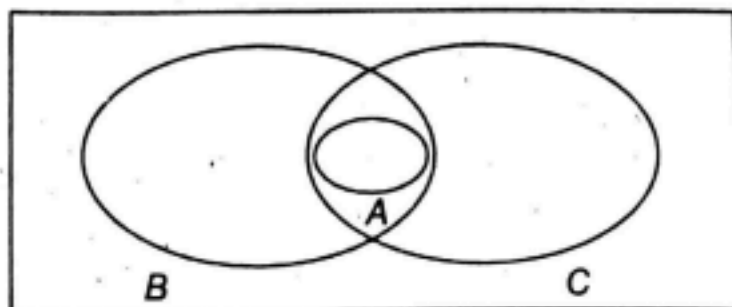
Q.9 Use a Venn diagram to illustrate the set of all months of the year whose names do not contain the letter R in the set of all months of the year.

Sol:



Q.10 Use a Venn diagram to illustrate the relationships $A \subset B$ and $A \subset C$.

Sol:



Q.11 For the following pairs of sets, determine whether the first is a subset of the second, the second is a subset of the first, or neither is a subset of the other.

- (a) the set of airline flights from Islamabad to Chicago, the set of nonstop airline flights from Islamabad to Chicago
- (b) the set of people who speak English, the set of people who speak Chinese
- (c) the set of flying squirrels, the set of living creatures that can fly

Sol:(a) Let

A = the set of airline flights from Islamabad to Chicago

B = the set of nonstop airline flights from Islamabad to Chicago

Obviously, $B \subseteq A$

(b) Let

A = the set of people who speak English

B = the set of people who speak Chinese

Obviously, neither $B \subseteq A$ nor $A \subseteq B$

(c) Let

A = the set of flying squirrels

B = the set of living creatures that can fly

Obviously, $A \subseteq B$

EXERCISE 2.2

Multiple Choice Questions (MCQs)

Q.1 Tick ✓ the correct option in the following questions:

(i) $A \cup B =$

(a) $\{x : x \in A \wedge x \in B\}$

(b) $\{x : x \in A \vee x \in B\}$

(c) $\exists x(x \in A \leftrightarrow x \in B)$

(d) $\forall x(x \in A \leftrightarrow x \in B)$

Sol: Correct answer is option (b).

(ii) $A \cap B =$

(a) $\{x : x \in A \wedge x \in B\}$

(b) $\{x : x \in A \vee x \in B\}$

(c) $\exists x(x \in A \leftrightarrow x \in B)$

(d) $\forall x(x \in A \leftrightarrow x \in B)$

Sol: Correct answer is option (a).

(iii) $A - B =$

(a) $\{x : x \in A \wedge x \in B\}$

(b) $\{x : x \in A \vee x \in B\}$

(c) $\{x : x \in A \wedge x \notin B\}$

(d) $\{x : x \in B \wedge x \notin A\}$

Sol: Correct answer is option (c).

(iii) $A - B =$

(a) $\{x : x \in A \wedge x \in B\}$

(b) $\{x : x \in A \vee x \in B\}$

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- Sol:** (c) $\{x : x \in A \wedge x \notin B\}$ (d) $\{x : x \in B \wedge x \notin A\}$
 Correct answer is option (c).
(iv) $B - A =$
- (a) $\{x : x \in A \wedge x \in B\}$ (b) $\{x : x \in A \vee x \in B\}$
 (c) $\{x : x \in A \wedge x \notin B\}$ (d) $\{x : x \in B \wedge x \notin A\}$
- Sol:** Correct answer is option (d).
(v) $A^c =$
- (a) $\{x \in U : x \notin A^c\}$ (b) $\{x \in U : x \notin A\}$
 (c) $\{x \in U : x \in A\}$ (d) none of these
- Sol:** Correct answer is option (a).
(vi) $A - B =$
- (a) $A \cap B^c$ (b) $A^c \cap B$
 (c) $A \cup B^c$ (d) none of these
- Sol:** Correct answer is option (a).
(vii) $A \cap U =$
- (a) U (b) A
 (c) $\emptyset$ (d) A^c
- Sol:** Correct answer is option (b).
(viii) $A \cup \emptyset =$
- (a) U (b) A
 (c) $\emptyset$ (d) A^c
- Sol:** Correct answer is option (b).
(ix) $A \cup U =$
- (a) U (b) A
 (c) $\emptyset$ (d) A^c
- Sol:** Correct answer is option (a).
(x) $A \cap \emptyset =$
- (a) U (b) A
 (c) $\emptyset$ (d) A^c
- Sol:** Correct answer is option (c).
(xi) Let $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$, then the bit string of the set $A = \{4, 8\}$ is
- (a) 0001 0001 00 (b) 0101 0001 01
 (c) 1100 0001 10 (d) 1110 0001 00
- Sol:** Correct answer is option (a).
(xii) Let $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$, then the bit string of the set $\{2, 4, 8, 10\}$ is
- (a) 0001 0001 00 (b) 0101 0001 01
 (c) 1100 0001 10 (d) 1110 0001 00
- Sol:** Correct answer is option (b).
(xiii) Let $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$, then the bit string of the set $\{1, 2, 8, 9\}$ is
- (a) 0001 0001 00 (b) 0101 0001 01
 (c) 1100 0001 10 (d) 1110 0001 00
- Sol:** Correct answer is option (c).

- (xiv) Let $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$, then the bit string of the set $\{1, 2, 3, 8\}$ is

(a) 0001 0001 00 (b) 0101 0001 01
(c) 1100 0001 10 (d) 1110 0001 00

Sol: Correct answer is option (d).

- (xv) Let $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$, then the bit string of the intersection of sets $\{1, 2, 3, 8\}$ and $\{4, 5, 6, 7\}$ is

(a) 0001 0001 00 (b) 0101 0001 01
(c) 0000 0000 00 (d) 1110 0001 00

Sol: $\{1, 2, 3, 8\} \cap \{4, 5, 6, 7\} = \phi$
Correct answer is option (c).

- (xvi) Let $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$, then the bit string of the union of sets $\{1, 2, 3, 8\}$ and $\{4, 5, 6, 7\}$ is

(a) 1111 1111 00 (b) 0101 0001 01
(c) 0000 0000 00 (d) 1110 0001 00

Sol: $\{1, 2, 3, 8\} \cup \{4, 5, 6, 7\} = \{1, 2, 3, 4, 5, 6, 7, 8\}$
Correct answer is option (a).

- (xvii) Symmetric difference of sets $\{1, 2, 3, 8\}$ and $\{4, 5, 6, 7\}$ is

(a) $\{1, 2, 3, 8\}$ (b) $\{4, 5, 6, 7\}$
(c) $\{1, 2, 3, 4, 5, 6, 7, 8\}$ (d) $\emptyset$

Sol: $\{1, 2, 3, 8\} \oplus \{4, 5, 6, 7\} = \{1, 2, 3, 4, 5, 6, 7, 8\}$
Correct answer is option (c).

- (xviii) If A is a set of all English alphabets, then the number of elements in its successor is

(a) 25 (b) 26
(c) 27 (d) 28

Sol: A contains 26 elements, so its successor contains 27 elements.
Correct answer is option (c).

Short Questions

Q.2 Solve / write answers of the following short questions:

- (i) Let A and B be sets. Show that $(A \cap B) \subseteq A$.

Sol: Let

$$\begin{aligned} x &\in (A \cap B) \\ x &\in A \text{ and } x \in B \\ \Rightarrow (A \cap B) &\subseteq A \end{aligned}$$

- (ii) Let A and B be sets. Show that $A \subseteq (A \cup B)$.

Sol: Let

$$\begin{aligned} x &\in A \\ x &\in (A \cup B) \\ \Rightarrow A &\subseteq A \cup B \end{aligned}$$

- (iii) Let A and B be sets. Show that $A - B \subseteq A$.

Sol: Let $x \in A - B$
 $x \in A$ and $x \notin B$
 $\Rightarrow A - B \subseteq A$

(iv) Let A and B be sets. Show that $A \cap (B - A) = \emptyset$.

Sol: Let $x \in A \cap (B - A)$
 $x \in A$ and $x \in (B - A)$
 $x \in A$ and $x \in B$ and $x \notin A$

Which is not possible, so $A \cap (B - A) = \emptyset$.

(v) Let A and B be sets. Show that $A \cup (B - A) = A \cup B$.

Sol: Let $x \in A \cup (B - A)$
 $x \in A$ or $x \in (B - A)$
 $x \in A$ or $x \in B$ and $x \notin A$
 $x \in A \cup B$
 $\Rightarrow A \cup (B - A) \subseteq A \cup B$ (1)

Conversely, let $x \in A \cup B$

$x \in A$ or $x \in B$
 $x \in A$ or $x \in B$ and $x \notin A$
 $x \in A \cup (B - A)$

$\Rightarrow A \cup B \subseteq A \cup (B - A)$ (2)

Combining (1) and (2), we have $A \cup (B - A) = A \cup B$.

(vi) If $A \subseteq B$, then show that $A \cap B = A$.

Sol: $A \subseteq B$
 $\Rightarrow A \cap B \subseteq B \cap B$
 $\Rightarrow A \cap B \subseteq B$ $\because B \cap B = B$

(vii) If $A \subseteq B$, then show that $A \cup B = B$.

Sol: $A \subseteq B$
 $\Rightarrow A \cup B \subseteq B \cup B$
 $\Rightarrow A \cup B \subseteq B$ (1) $\because B \cup B = B$

Also $B \subseteq A \cup B$ (2)

Combining (1) and (2), we have $A \cup B = B$.

(viii) If A is a subset of a universal set U , then show that $A \oplus A = \emptyset$.

Sol: It is clear from the definition of symmetric difference of two sets that $A \oplus A = \emptyset$.

(ix) If A is a subset of a universal set U , then show that $A \oplus \emptyset = A$.

Sol: Let		Conversely, let
$x \in A \oplus \emptyset$		$x \in A$
$\Rightarrow x \in A$		$\Rightarrow x \in A \oplus \emptyset$
$\Rightarrow A \oplus \emptyset \subseteq A$	(1)	$\Rightarrow A \subseteq A \oplus \emptyset$
		(2)

Combining (1) and (2), we have $A \oplus \emptyset = A$.

(x) If A is a subset of a universal set U , then show that $A \oplus U = A^c$.

Sol: Let

$$\begin{aligned} x &\in A \oplus U \\ \Rightarrow x &\in U \text{ and } x \notin A \\ \Rightarrow x &\in U - A \\ \Rightarrow x &\in A^c \\ \Rightarrow A \oplus U &\subseteq A^c \quad (1) \end{aligned}$$

Conversely, let

$$\begin{aligned} x &\in A^c \\ \Rightarrow x &\in U \text{ and } x \notin A \\ \Rightarrow x &\in U - A \\ \Rightarrow x &\in A \oplus U \\ \Rightarrow A^c &\subseteq A \oplus U \quad (2) \end{aligned}$$

Combining (1) and (2), we have $A \oplus U = A^c$.

(xi) If A is a subset of a universal set U , then show that $A \oplus A^c = U$.

Sol: Let

$$\begin{aligned} x &\in A \oplus A^c \\ \Rightarrow x &\in U \\ \Rightarrow A \oplus A^c &\subseteq U \quad (1) \end{aligned}$$

Conversely, let

$$\begin{aligned} x &\in U \\ \Rightarrow x &\in A \oplus A^c \\ \Rightarrow U &\subseteq A \oplus A^c \quad (2) \end{aligned}$$

Combining (1) and (2), we have $A \oplus A^c = U$.

(xii) Find the symmetric difference of N and Z .

Sol: $N \oplus Z = \{ \dots, -3, -2, -1, 0 \}$

(xiii) What is the successor of $\{a, b, c\}$?

Sol: The successor of $\{a, b, c\}$ is $\{a, b, c, \{a, b, c\}\}$

(xiv) If $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ is the universal set, then find the bit string of $\{1, 6, 9\}$.

Sol: The bit string of $\{1, 6, 9\}$ is 1000 0100 10

(xv) If $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ is the universal set, then find the set with bit string 1100 0011 01.

Sol: The set with bit string 1100 0011 01 is $\{1, 2, 7, 8, 10\}$

(xvi) What subset of a finite universal set does the string with all zeros represent?

Sol: The string with all zeros represents the empty subset of finite universal set.

(xvii) What subset of a finite universal set does the string with all ones represent?

Sol: The string with all ones represents the universal set itself.

Long Questions

Q.3 Let A and B be subsets of a universal set U . Show that $A \subseteq B$ if and only if $B^c \subseteq A^c$.

Sol: Let $A \subseteq B$

$$x \in B^c$$

$$x \notin B$$

$$x \notin A \quad \because A \subseteq B$$

$$x \in A^c$$

$$\Rightarrow B^c \subseteq A^c \quad (1)$$

Let $B^c \subseteq A^c$

$$x \in A$$

$$x \notin A^c$$

$$x \notin B^c \quad \because B^c \subseteq A^c$$

$$x \in AB$$

$$\Rightarrow A \subseteq B \quad (2)$$

This shows that $A \subseteq B$ if and only if $B^c \subseteq A^c$.

Q.4

Sol:

Let $A \oplus B = (A \cup B) - (A \cap B)$.

Let

$$x \in A \oplus B$$

$$x \in A \text{ and } x \notin B \text{ or } x \notin A \text{ and } x \in B$$

$$x \in A \text{ or } x \in B \text{ and } x \notin B \text{ or } x \notin A$$

$$x \in A \cup B \text{ and } x \notin (A \cap B)$$

$$x \in (A \cup B) - (A \cap B)$$

$$\Rightarrow A \oplus B \subseteq (A \cup B) - (A \cap B) \quad (1)$$

Conversely, let

$$x \in (A \cup B) - (A \cap B)$$

$$x \in (A \cup B) \text{ and } x \notin (A \cap B)$$

$$x \in A \text{ or } x \in B \text{ and } x \notin A \text{ or } x \notin B$$

$$x \in A \text{ and } x \notin B \text{ or } x \in B \text{ and } x \notin A$$

$$x \in A \oplus B$$

$$\Rightarrow (A \cup B) - (A \cap B) \subseteq A \oplus B \quad (2)$$

Combining (1) and (2), we have

$$A \oplus B = (A \cup B) - (A \cap B)$$

Q.5

Sol:

Let $A \oplus B = (A - B) \cup (B - A)$.

Let

$$x \in A \oplus B$$

$$x \in A \text{ and } x \notin B \text{ or } x \notin A \text{ and } x \in B$$

$$x \in A - B \text{ or } x \in B - A$$

$$x \in (A - B) \cup (B - A)$$

$$\Rightarrow A \oplus B \subseteq (A - B) \cup (B - A) \quad (1)$$

Conversely, let $x \in (A - B) \cup (B - A)$

$$x \in (A - B) \text{ or } x \in (B - A)$$

$$x \in A \text{ and } x \notin B \text{ or } x \in B \text{ and } x \notin A$$

$$x \in A \oplus B$$

$$\Rightarrow (A - B) \cup (B - A) \subseteq A \oplus B \quad (2)$$

Combining (1) and (2), we have

$$A \oplus B = (A - B) \cup (B - A)$$

Q.6 Find the successors of the following sets.

(a) $\{1, 2, 3\}$

(b) $\emptyset$

(c) $\{\emptyset\}$

(d) $\{\emptyset, \{\emptyset\}\}$

Sol:(a) The successor of $\{1, 2, 3\}$ is $\{1, 2, 3, \{1, 2, 3\}\}$.

(b) The successor of $\emptyset$ is $\{\emptyset\}$.

(c) The successor of $\{\emptyset\}$ is $\{\emptyset, \{\emptyset\}\}$.

(d) The successor of $\{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}\}$.

Q.7 Find the successors of the following sets.

(a) $\{2, 3\}$

(b) $\{1, 3\}$

(c) $\{1\}$

(d) $\{1, \{1\}\}$

Sol:(a) The successor of $\{2, 3\}$ is $\{2, 3, \{2, 3\}\}$.

(b) The successor of $\{1, 3\}$ is $\{1, 3, \{1, 3\}\}$.

(c) The successor of $\{1\}$ is $\{1, \{1\}\}$.

(d) The successor of $\{1, \{1\}\}$ is $\{1, \{1\}, \{1, \{1\}\}\}$.

Q.8 Using the universal set $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$, find the set specified by each of these bit strings.

(a) 1000 0000 01

(b) 1111 0011 11

(c) 0101 1110 00

(d) 0111 1110 01

Sol:(a) The set specified by the bit string 1000 0000 01 is $\{1, 10\}$

(b) The set specified by the bit string 1111 0011 11 is $\{1, 2, 3, 4, 7, 8, 9, 10\}$

(c) The set specified by the bit string 0101 1110 00 is $\{2, 4, 5, 6, 7\}$

(d) The set specified by the bit string 0111 1110 01 is $\{2, 3, 4, 5, 6, 7, 10\}$

Q.9 Suppose that the universal set is $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$. Express each of the following sets with bit strings where the i th bit in the string is 1 if i is in the set and 0 otherwise.

(a) $\{2, 3, 4, 5\}$

(b) $\{1, 3, 6, 9\}$

(c) $\{1, 4, 7, 8, 9\}$

(d) $\{3, 4, 6, 9, 10\}$

Sol:(a) The bit string of the set $\{2, 3, 4, 5\}$ is 0111 1000 00

(b) The bit string of the set $\{1, 3, 6, 9\}$ is 1010 0100 10

(c) The bit string of the set $\{1, 4, 7, 8, 9\}$ is 1001 0011 10

(d) The bit string of the set $\{3, 4, 6, 9, 10\}$ is 0011 0100 11

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Q.10 Find $\bigcup_{i=1}^{\infty} A_i$ and $\bigcap_{i=1}^{\infty} A_i$ if for every positive integer i ,

(a) $A_i = \{-i, -i+1, \dots, -1, 0, 1, \dots, i-1, i\}$

Sol: $A_i = \{-i, -i+1, \dots, -1, 0, 1, \dots, i-1, i\}$

From this set, we have

$$A_1 = \{-1, 0, 1\}$$

$$A_2 = \{-2, -1, 0, 1, 2\}$$

$$A_3 = \{-3, -2, -1, 0, 1, 2, 3\}$$

Therefore

$$\begin{aligned}\bigcup_{i=1}^{\infty} A_i &= A_1 \cup A_2 \cup A_3 \cup \dots \\ &= \{0, \pm 1, \pm 2, \pm 3, \dots\} \\ &= \mathbb{Z}\end{aligned}$$

Similarly,

$$\begin{aligned}\bigcap_{i=1}^{\infty} A_i &= A_1 \cap A_2 \cap A_3 \cap \dots \\ &= \{0\}\end{aligned}$$

(b) $A_i = \{-i, i\}$

Sol: $A_i = \{-i, i\}$

From this set, we have

$$A_1 = \{-1, 1\}$$

$$A_2 = \{-2, 2\}$$

$$A_3 = \{-3, 3\}$$

Therefore

$$\begin{aligned}\bigcup_{i=1}^{\infty} A_i &= A_1 \cup A_2 \cup A_3 \cup \dots \\ &= \{\pm 1, \pm 2, \pm 3, \dots\}\end{aligned}$$

Similarly,

$$\begin{aligned}\bigcap_{i=1}^{\infty} A_i &= A_1 \cap A_2 \cap A_3 \cap \dots \\ &= \emptyset\end{aligned}$$

(c) $A_i = [-i, i]$

Sol: $A_i = [-i, i]$. From this set, we have

$$A_1 = [-1, 1]$$

$$A_2 = [-2, 2]$$

$$A_3 = [-3, 3]$$

Therefore

$$\begin{aligned}\bigcup_{i=1}^{\infty} A_i &= A_1 \cup A_2 \cup A_3 \cup \dots \\ &= (-\infty, \infty) \\ &= \mathbb{R}\end{aligned}$$

Similarly,

$$\bigcap_{i=1}^{\infty} A_i = A_1 \cap A_2 \cap A_3 \cap \dots = [-1, 1]$$

(d) $A_i = [i, \infty)$ Sol: $A_i = [i, \infty)$

From this set, we have

$$A_1 = [1, \infty)$$

$$A_2 = [2, \infty)$$

$$A_3 = [3, \infty)$$

Therefore

$$\begin{aligned}\bigcup_{i=1}^{\infty} A_i &= A_1 \cup A_2 \cup A_3 \cup \dots \\ &= [1, \infty)\end{aligned}$$

Similarly,

$$\begin{aligned}\bigcap_{i=1}^{\infty} A_i &= A_1 \cap A_2 \cap A_3 \cap \dots \\ &= \phi\end{aligned}$$

Q.11 Find the sets A and B if

$$A - B = \{1, 5, 7, 8\}, B - A = \{2, 10\} \text{ and } A \cap B = \{3, 6, 9\}$$

PU, 2011 (BS Math)

Sol: $A = \{1, 3, 5, 6, 7, 8, 9\}$
 $B = \{2, 3, 6, 9, 10\}$ **Q.12** List the numbers of these sets:(a) $\{x : x \text{ is a real number such that } x^2 = 1\}$ (b) $\{x : x \text{ is a +ve (positive) integer less than 12}\}$ (c) $\{x : x \text{ is a square of a nonnegative integer and } x < 100\}$ (d) $\{x : x \text{ is an integer such that } x^2 = 2\}$

PU, 2011 (BS Math)

Sol: (a) $\{-1, 1\}$ (b) $\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11\}$ (c) $\{0, 1, 4, 9, 16, 25, 36, 49, 64, 81\}$ (d) ϕ **Q.13** Prove that $(A \cup (B \cap C))^c = (C^c \cup B^c) \cap A^c$.

PU, 2012 (BS Math)

Sol: Let

$$x \in (A \cup (B \cap C))^c$$

$$x \notin (A \cup (B \cap C))$$

$$x \notin A \text{ and } x \notin (B \cap C)$$

$$x \notin A \text{ and } x \notin B \text{ or } x \notin C$$

$$x \in A^c \text{ and } x \in B^c \text{ or } x \in C^c$$

$$x \in A^c \text{ and } x \in (B^c \cup C^c)$$

$$x \in A^c \cap (B^c \cup C^c)$$

$$x \in (C^c \cup B^c) \cap A^c$$

$$\Rightarrow (A \cup (B \cap C))^c \subseteq (C^c \cup B^c) \cap A^c$$

(1)

Conversely, let

$$x \in (C^c \cup B^c) \cap A^c$$

$$x \in A^c \cap (B^c \cup C^c)$$

$$x \in A^c \text{ and } x \in (B^c \cup C^c)$$

$$x \in A^c \text{ and } x \in B^c \text{ or } x \in C^c$$

$$x \notin A \text{ and } x \notin B \text{ or } x \notin C$$

$$x \notin A \text{ and } x \notin (B \cap C)$$

$$x \notin (A \cup (B \cap C))$$

$$x \in (A \cup (B \cap C))^c$$

$$\Rightarrow (C^c \cup B^c) \cap A^c \subseteq (A \cup (B \cap C))^c \quad (2)$$

Combining (1) and (2), we have

$$(A \cup (B \cap C))^c = (C^c \cup B^c) \cap A^c$$

EXERCISE 2.3

Multiple Choice Questions (MCQs)

Q.1 Tick ✓ the correct option in the following questions:

- (i) $(f_1 + f_2)(x) =$
 (a) $f_1(x) + f_2(x)$ (b) $f_1(x) - f_2(x)$
 (c) $f_1(x)f_2(x)$ (d) none of these

Sol: $(f_1 + f_2)(x) = f_1(x) + f_2(x)$
 Correct answer is option (a).

- (ii) If $f_1(x) = x^2$, $f_2(x) = 2 - 5x^2$, then $f_1(x) + f_2(x) =$
 (a) $2 - 5x^2$ (b) $2 - 4x^2$
 (c) $5x^2 - 2$ (d) $4x^2 - 2$

Sol: $f_1(x) + f_2(x) = x^2 + (2 - 5x^2) = 2 - 4x^2$
 Correct answer is option (b).

- (iii) $(f_1 f_2)(x) =$
 (a) $f_1(x) + f_2(x)$ (b) $f_1(x) - f_2(x)$
 (c) $f_1(x)f_2(x)$ (d) none of these

Sol: $(f_1 f_2)(x) = f_1(x)f_2(x)$
 Correct answer is option (c).

- (iv) If $f_1(x) = x^2$, $f_2(x) = 2 - 5x^2$, then $f_1(x)f_2(x) =$
 (a) $2 - 5x^2$ (b) $2 - 4x^2$
 (c) $5x^2 - 2$ (d) $2x^2 - 5x^4$

Sol: $f_1(x)f_2(x) = x^2(2 - 5x^2) = 2x^2 - 5x^4$
 Correct answer is option (d).

- (v) If $S = \{-1, 0, 1\}$ and $f(x) = 2x + 1$, then $f(S) =$
 (a) $\{-1, 0, 1\}$ (b) $\{-1, 1, 3\}$
 (c) $\{1, 2, 3\}$ (d) $\{7, 6, 5\}$



















